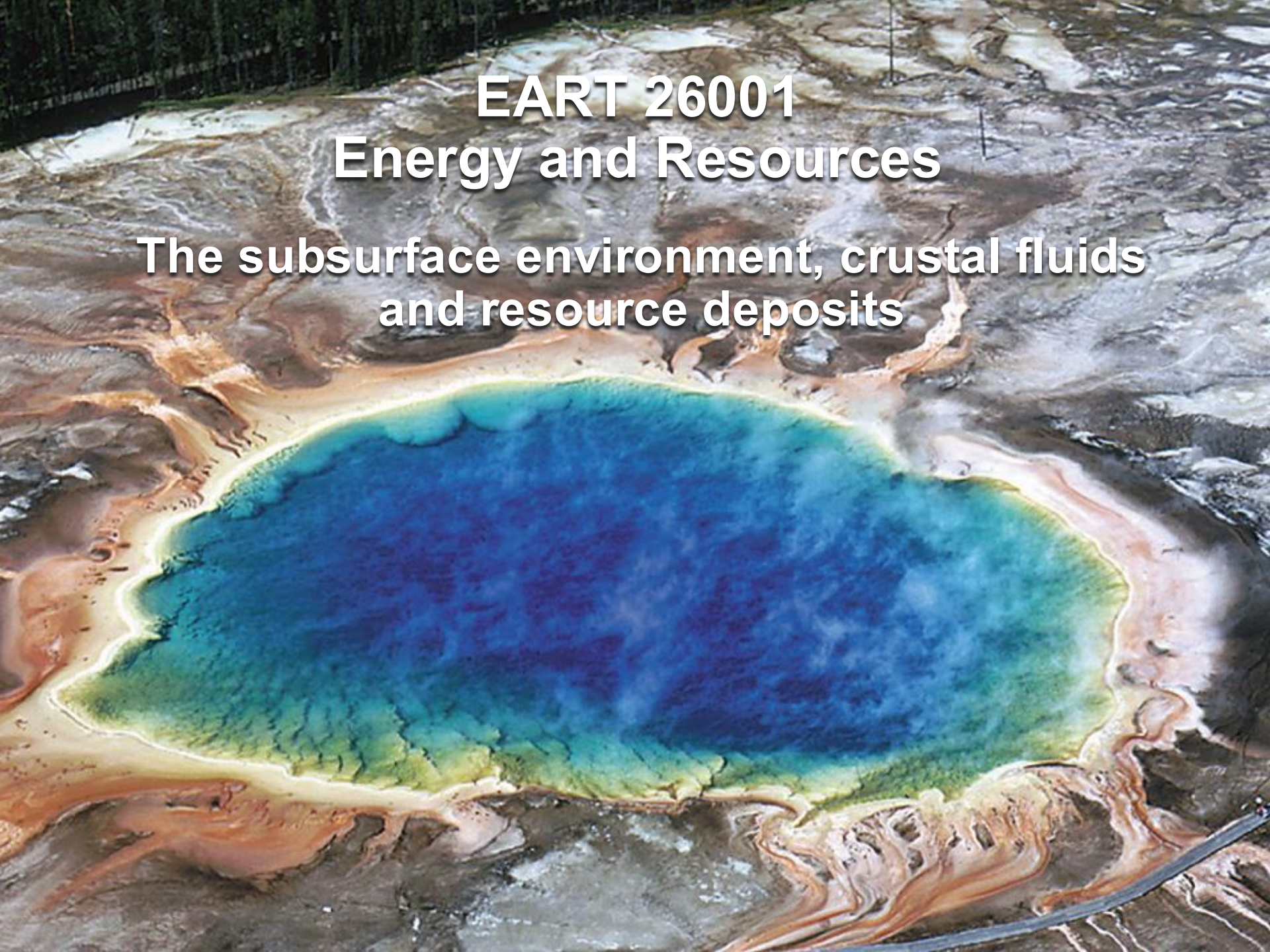


EART 26001
Energy and Resources

The subsurface environment, crustal fluids and resource deposits

Introduction
How to form a resource deposit



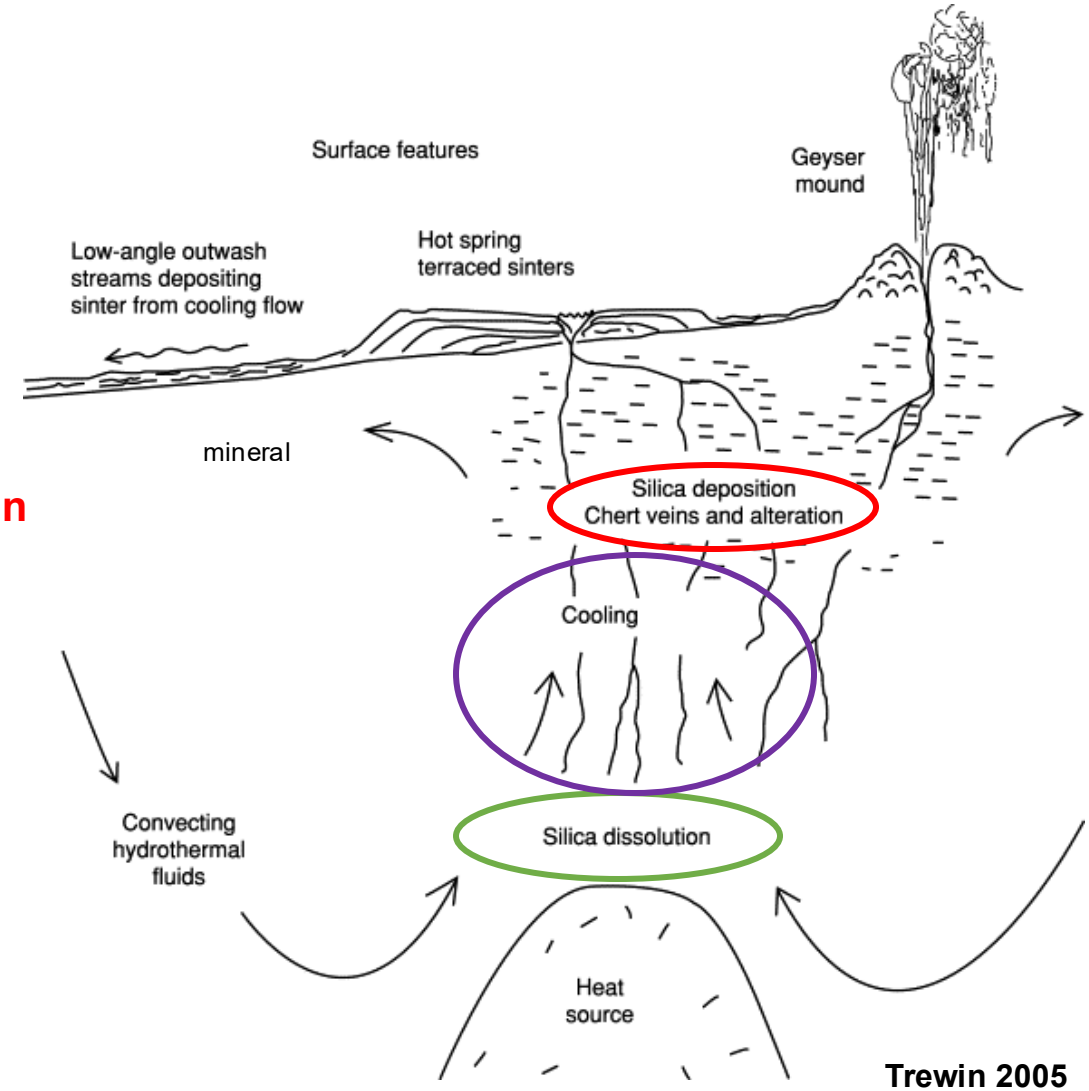
- Element enrichment/ transport?
- Transport pathway?
- Space for enrichment?
- Precipitation mechanism?
- **Subsurface properties, fluids and deposit types**

precipitation

transport

source

Schematic of a hydrothermal system

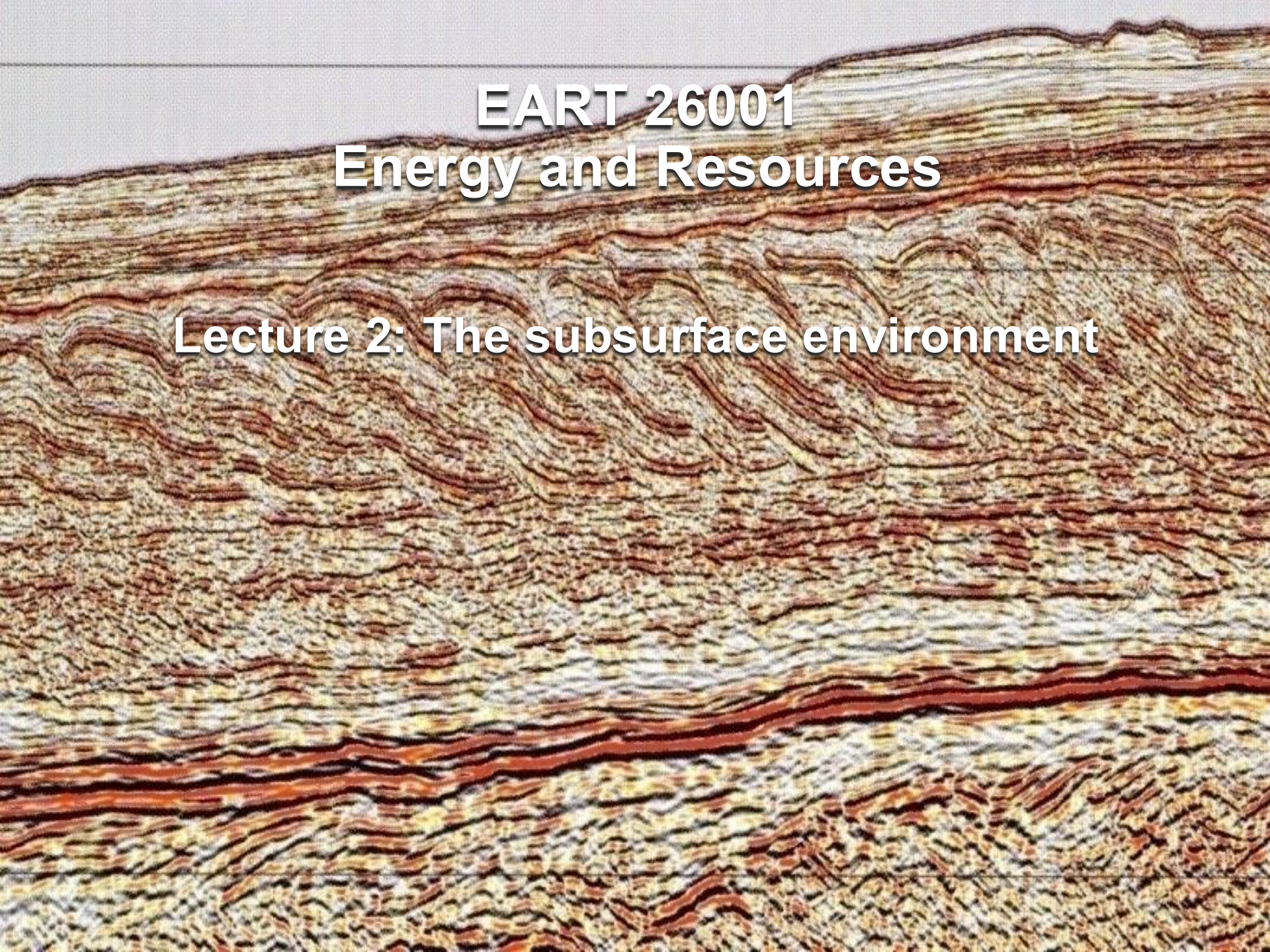


Trewin 2005

Energy and Resources
Learning goals weeks 2 and 3

In these lectures you will learn to:

- **Describe what fluids are involved in formation of energy and resource commodities, fluid origins and composition.**
- **Characterize physical conditions in the subsurface using porosity, permeability, thermal and pressure gradients, rock deformation, and how they control fluid circulation and commodity formation.**
- **Classify resource deposits in a simple scheme according to fluid type and site of deposit.**

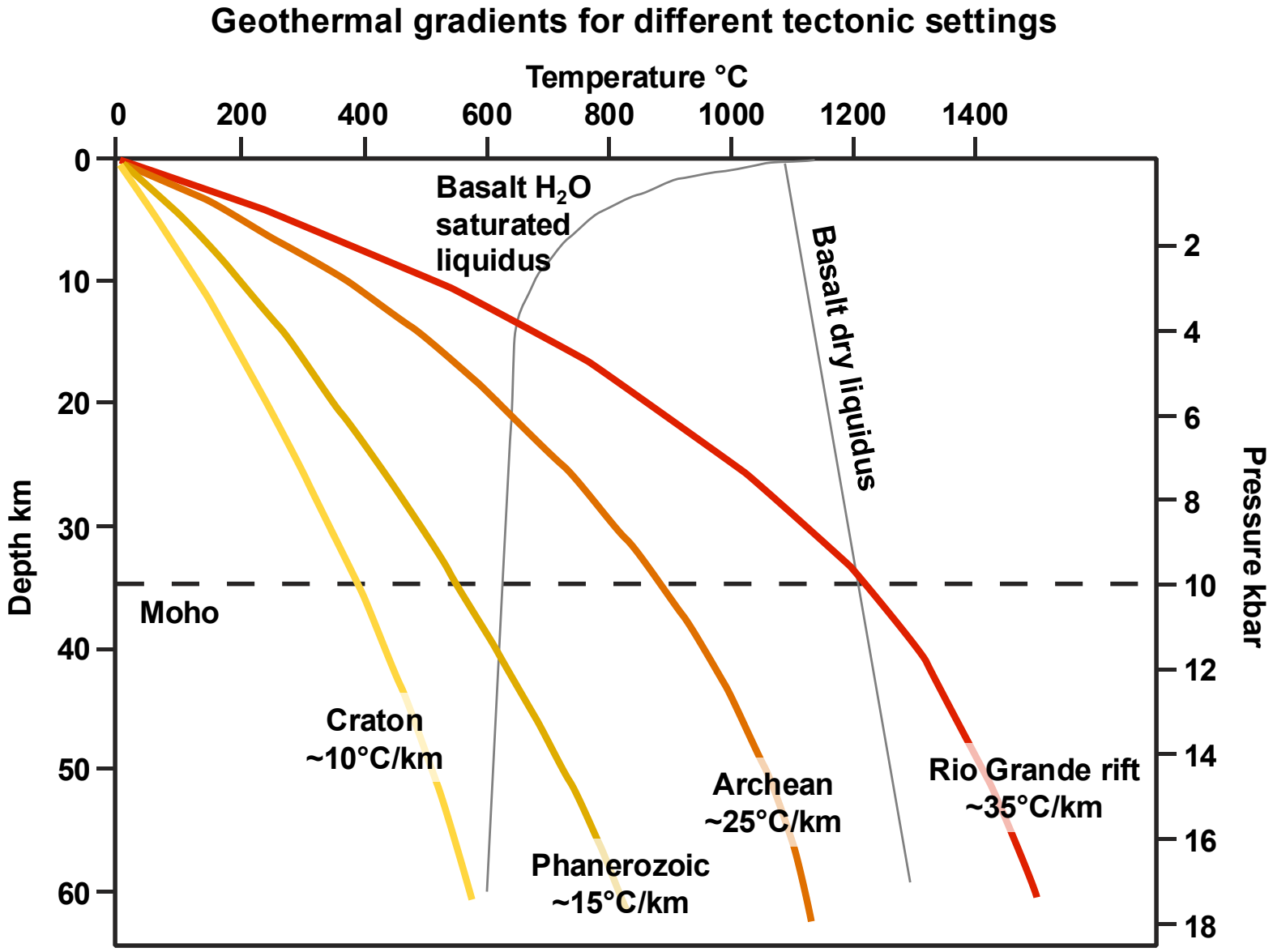


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Energy and Resources

Lecture 2: The subsurface environment

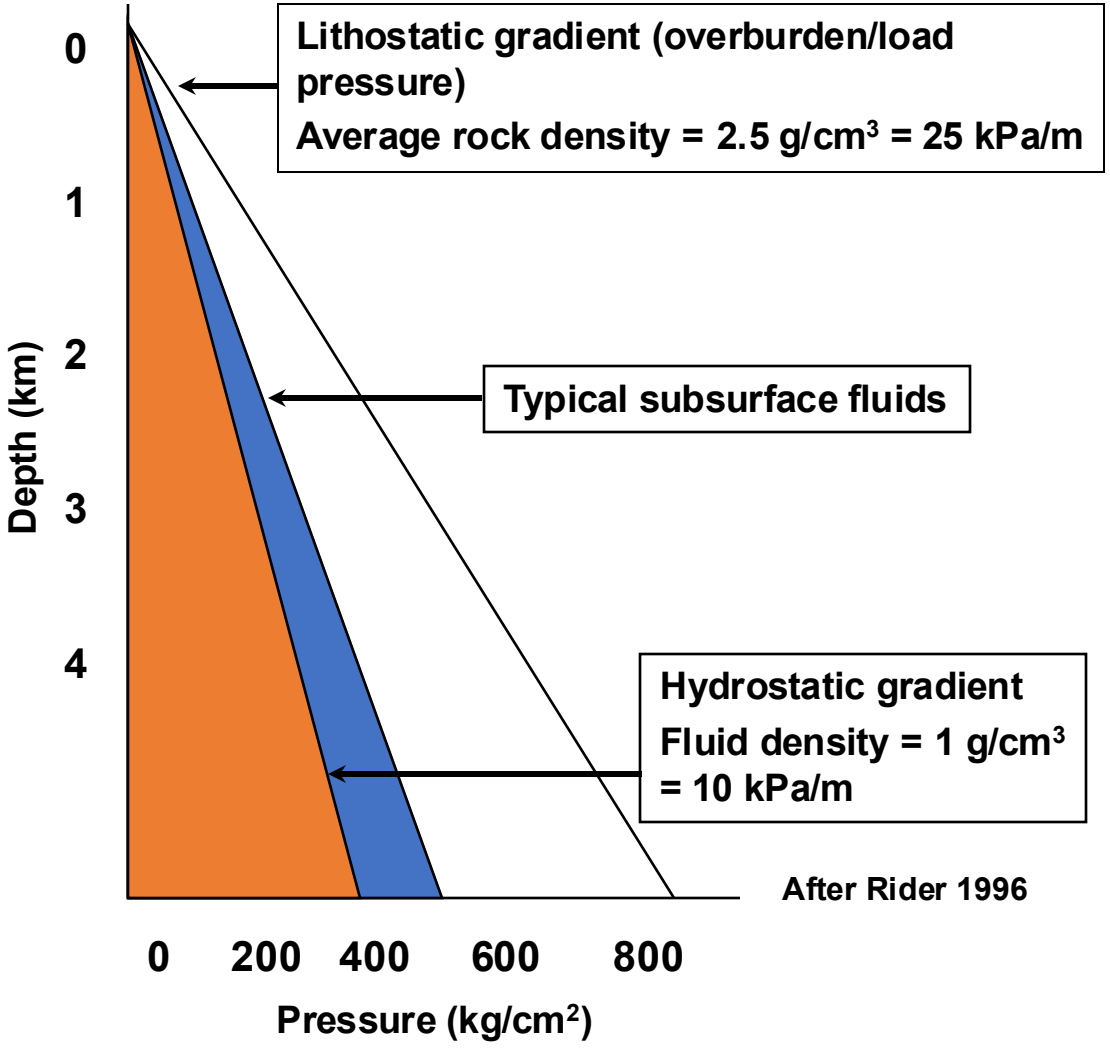
Key Subsurface Properties

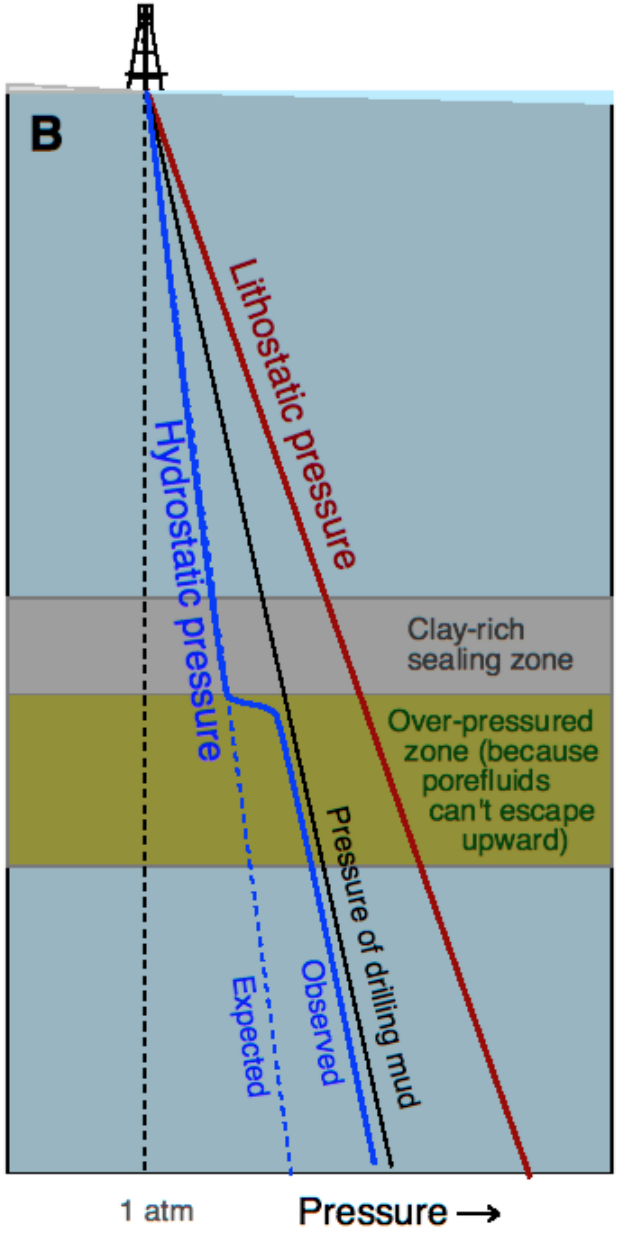
Temperature



Key Subsurface Properties
Pressure

- Weight of overlying rock exerts uniform load pressure = lithostatic stress/confining stress



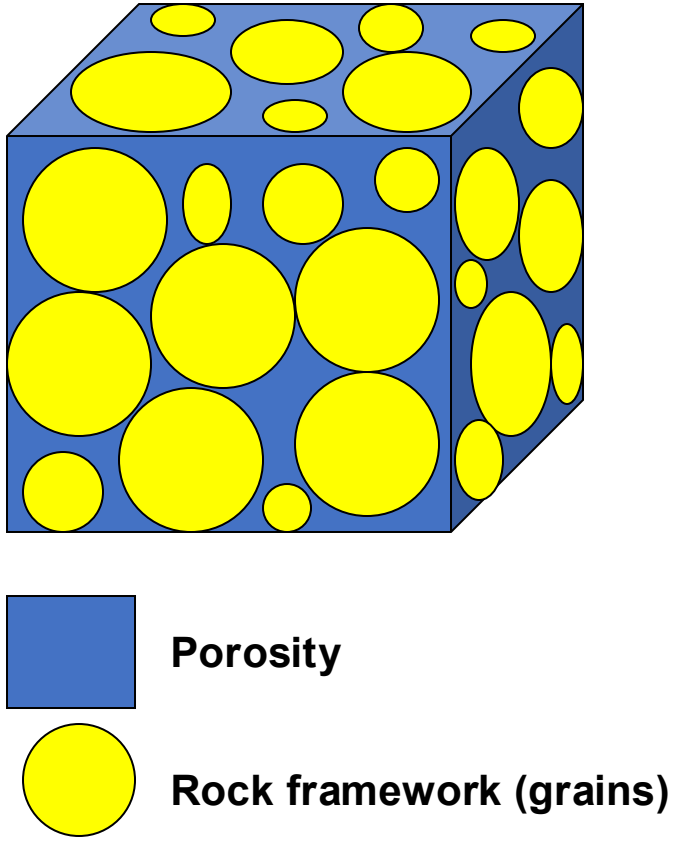


Source: Uni Georgia

Key Subsurface Properties
Porosity

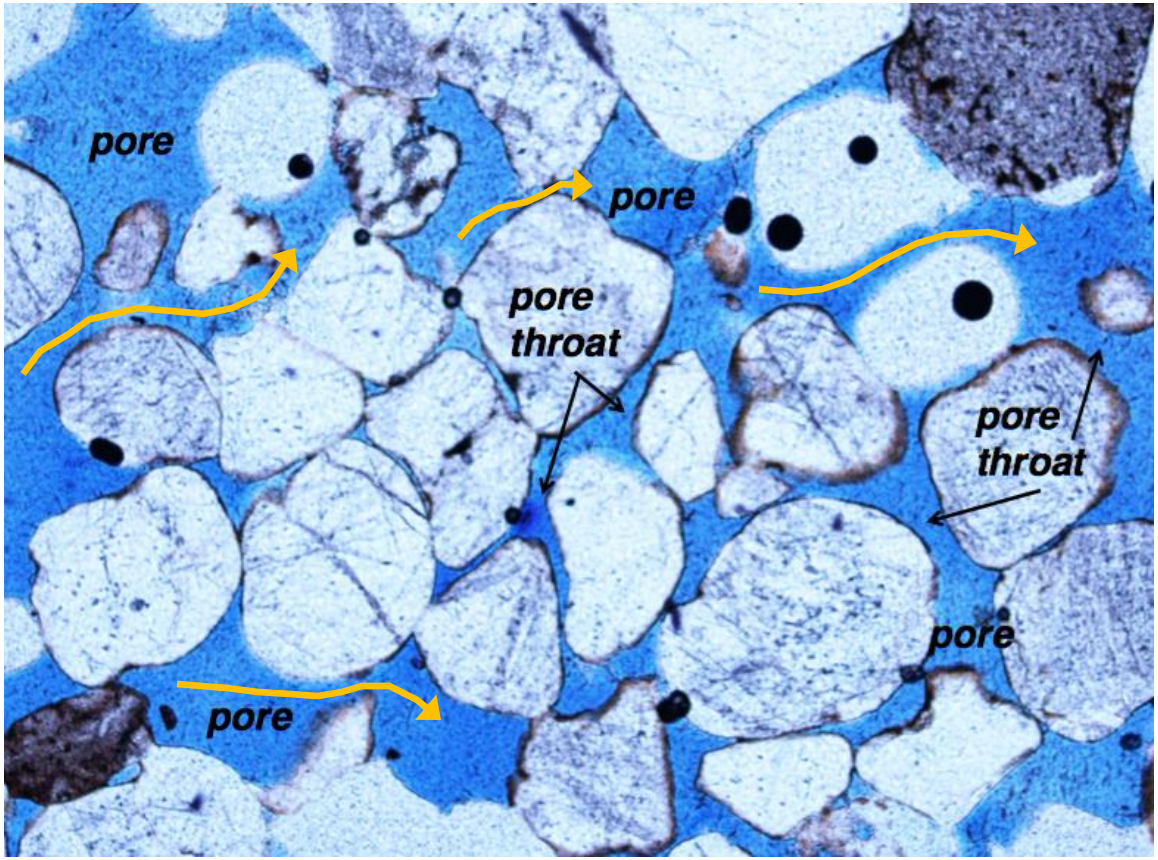
- Porosity is any space in a rock that can store fluid (water, oil, gas) – storage capacity of a rock
- Expressed as fraction or %

$$\text{Porosity } (\Phi) = \frac{\text{pore volume } (V_p)}{\text{total rock volume } (V_b)}$$

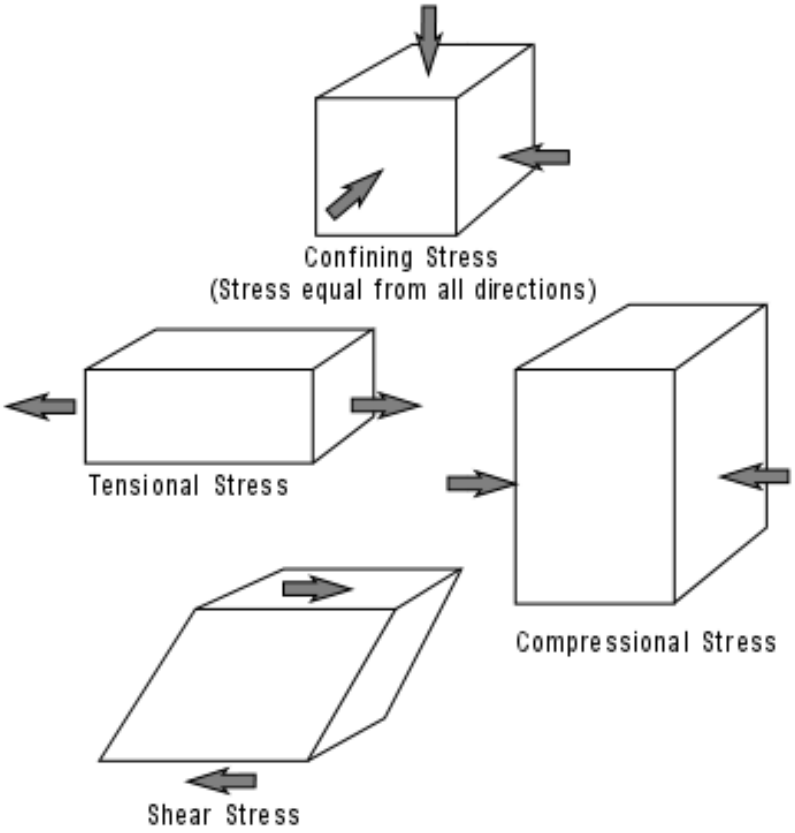


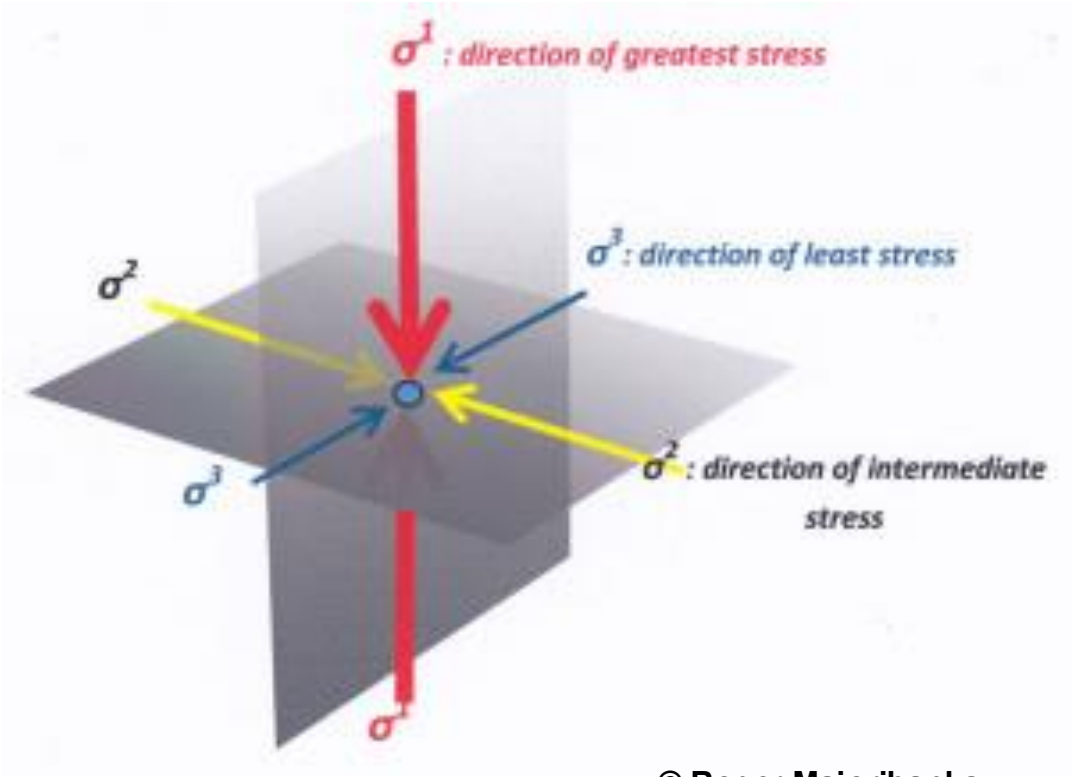
Key Subsurface Properties
Permeability

- Porous rocks have variable ability to flow fluids
- Permeability is the ability of a fluid to move through a rock, and is dependent on the type and connectivity of the pore network
- Expressed as m^2 or D (Darcy)



Key Subsurface Properties
Deformation of rocks

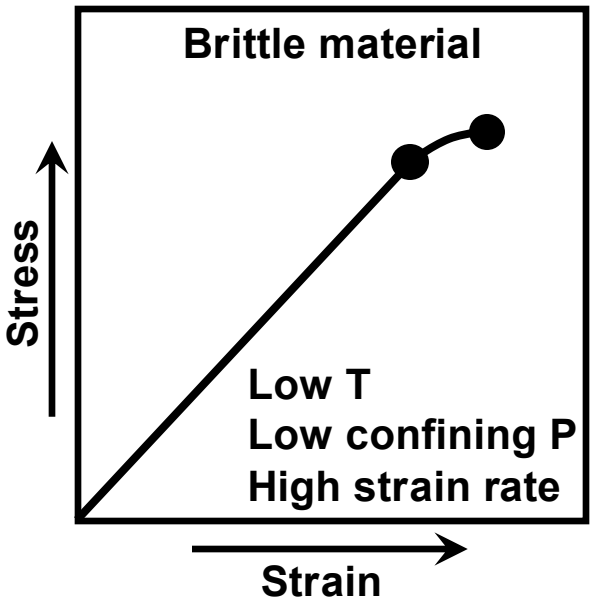


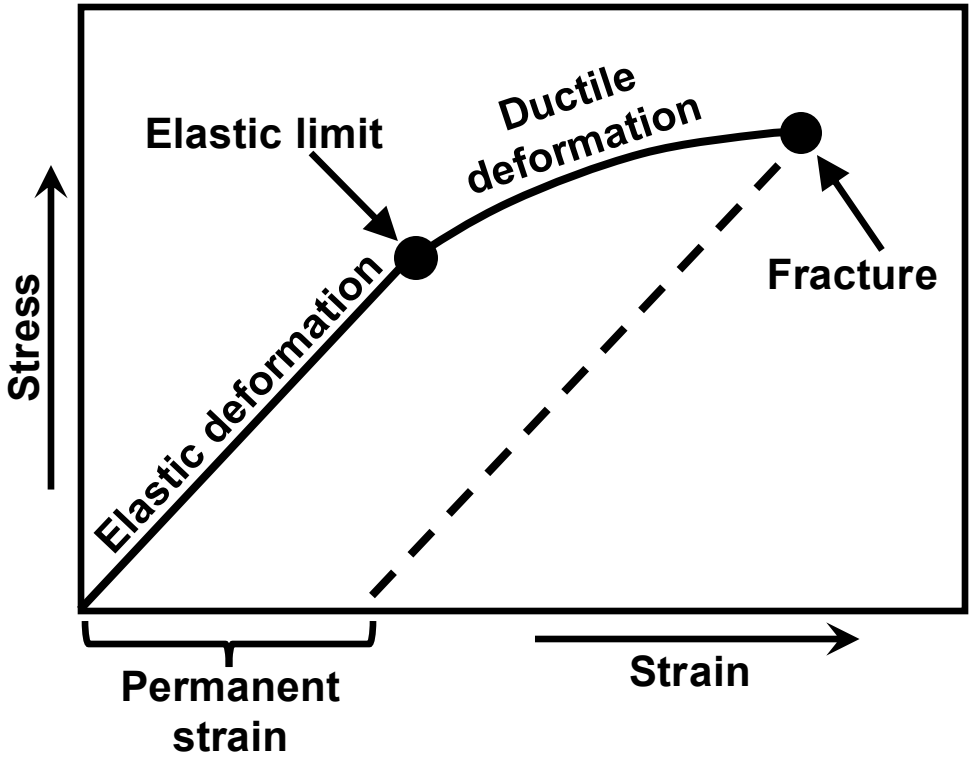


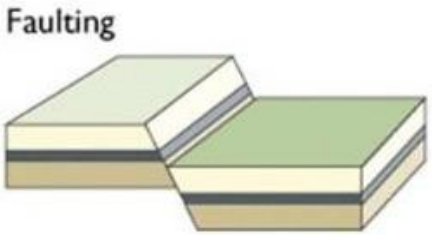
© Roger Majoribanks

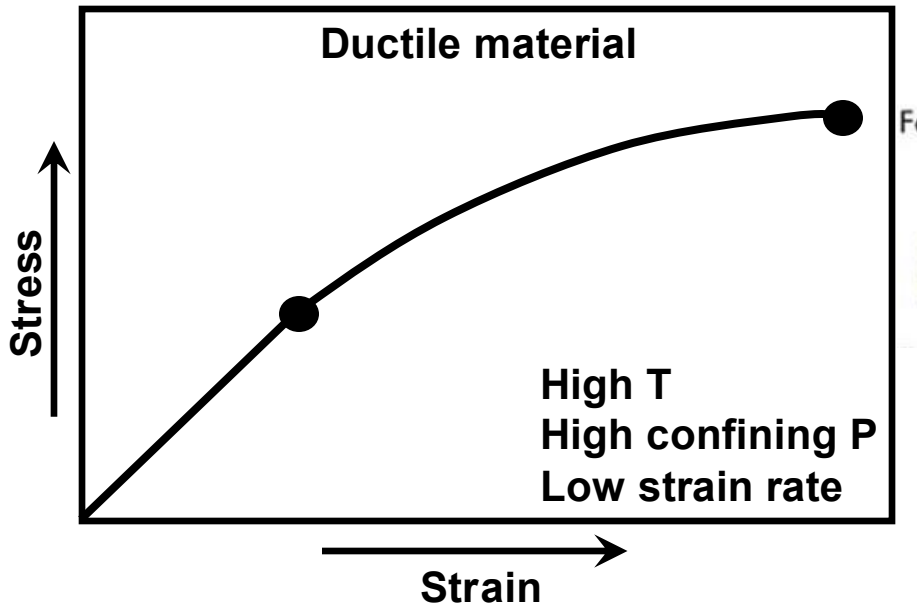
- Lateral transmission of tectonic forces through crust and gravity cause differential/ triaxial stress (stress = force applied per unit area)
- Triaxial stress: three principal stress vectors $\sigma 1 > \sigma 2 > \sigma 3$
- Reaction to stress = strain

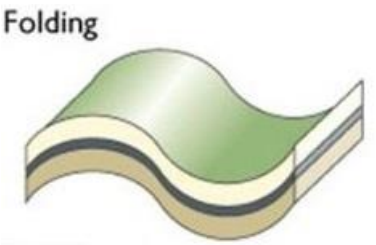
Key Subsurface Properties
Deformation of rocks



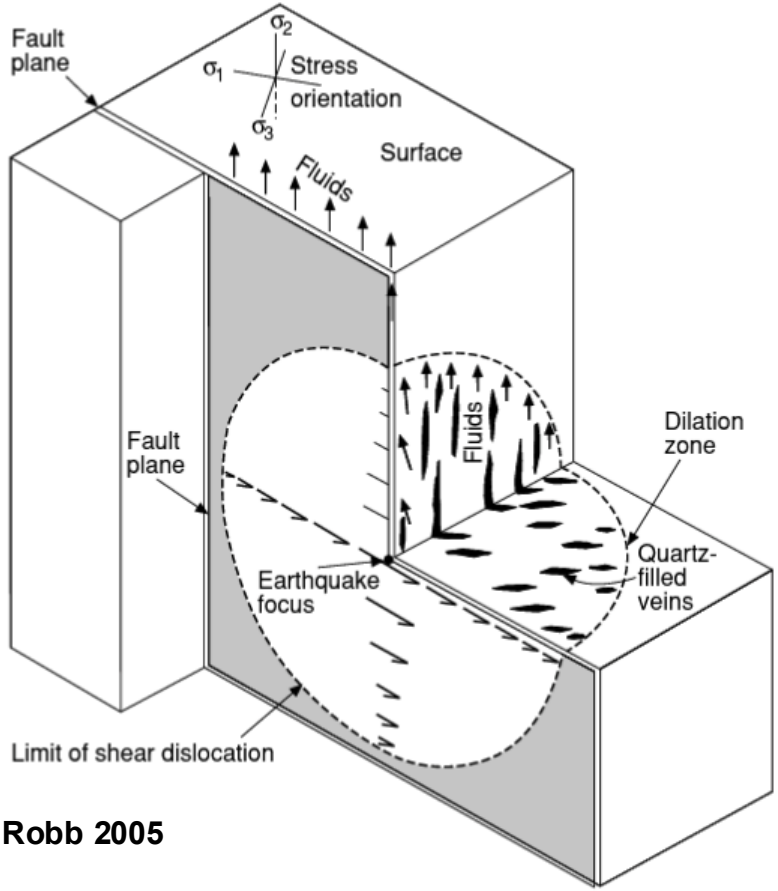




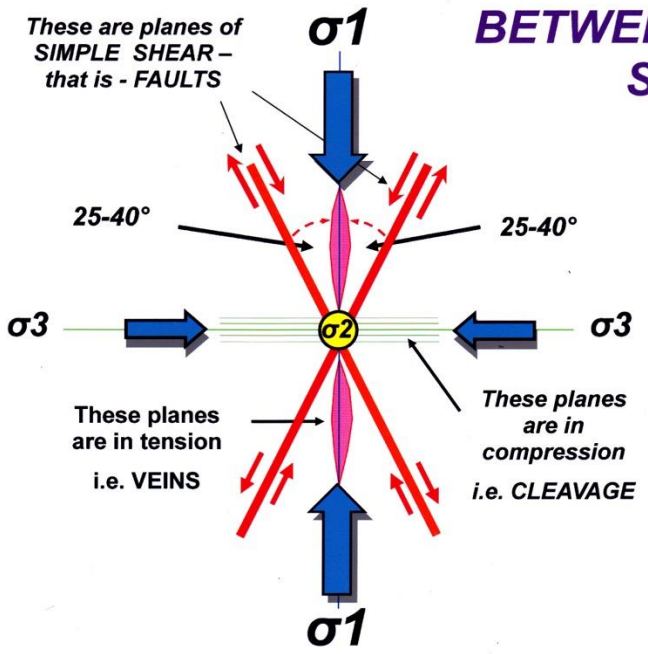




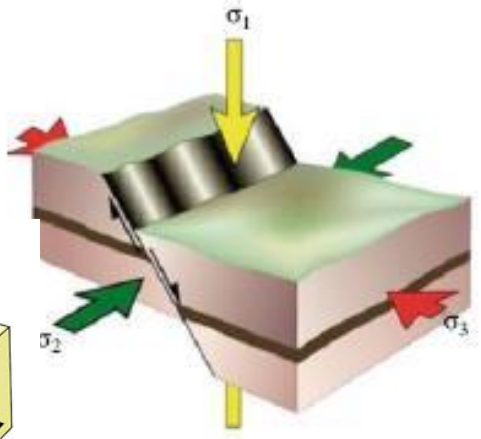
Key Subsurface Properties
Deformation of rocks

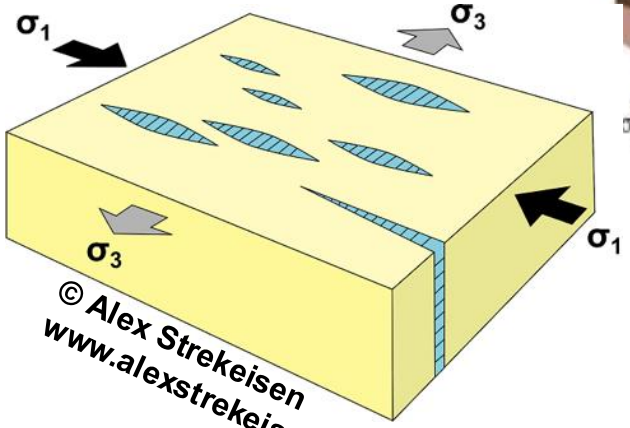


Robb 2005



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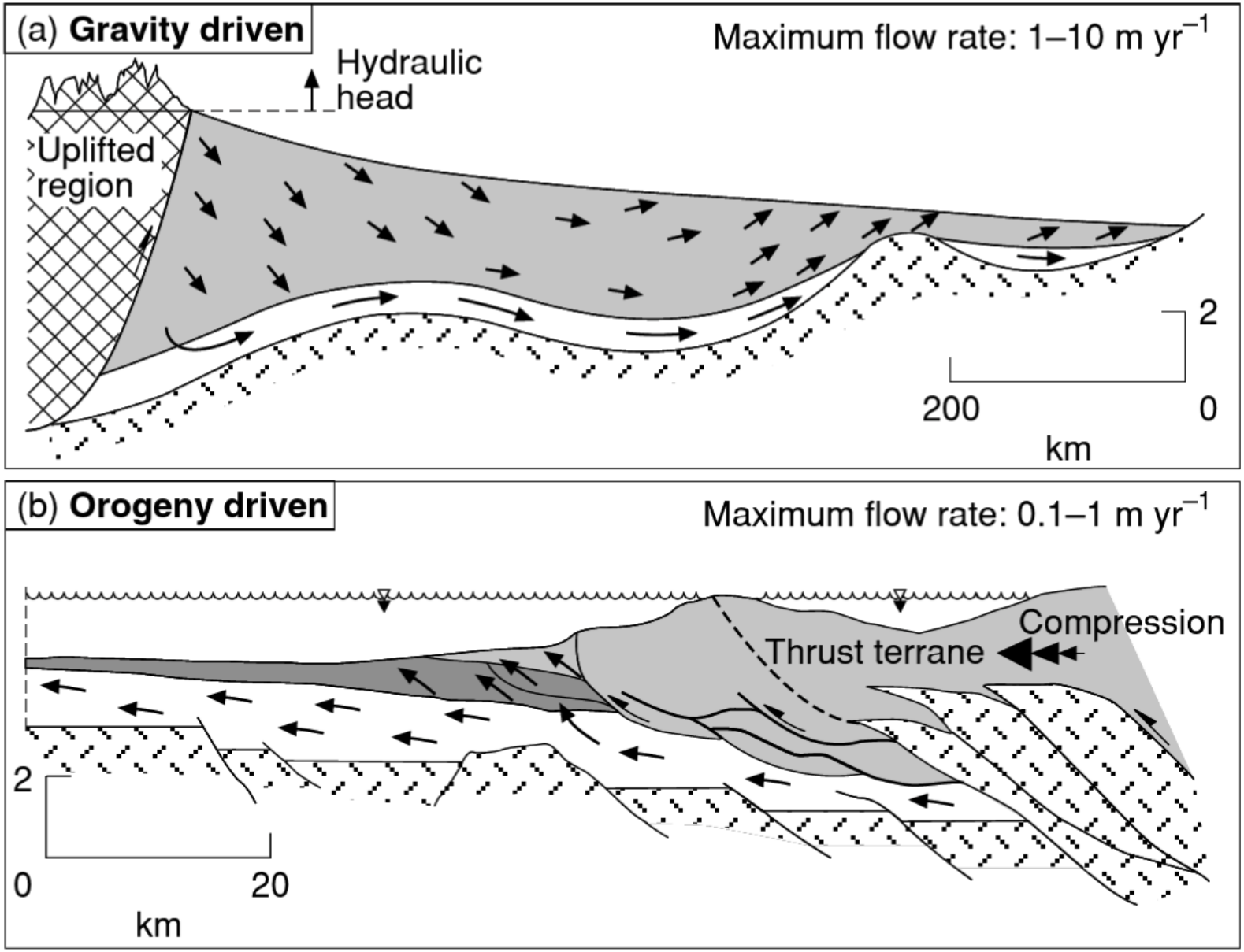




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www.alexstrekeisen.it

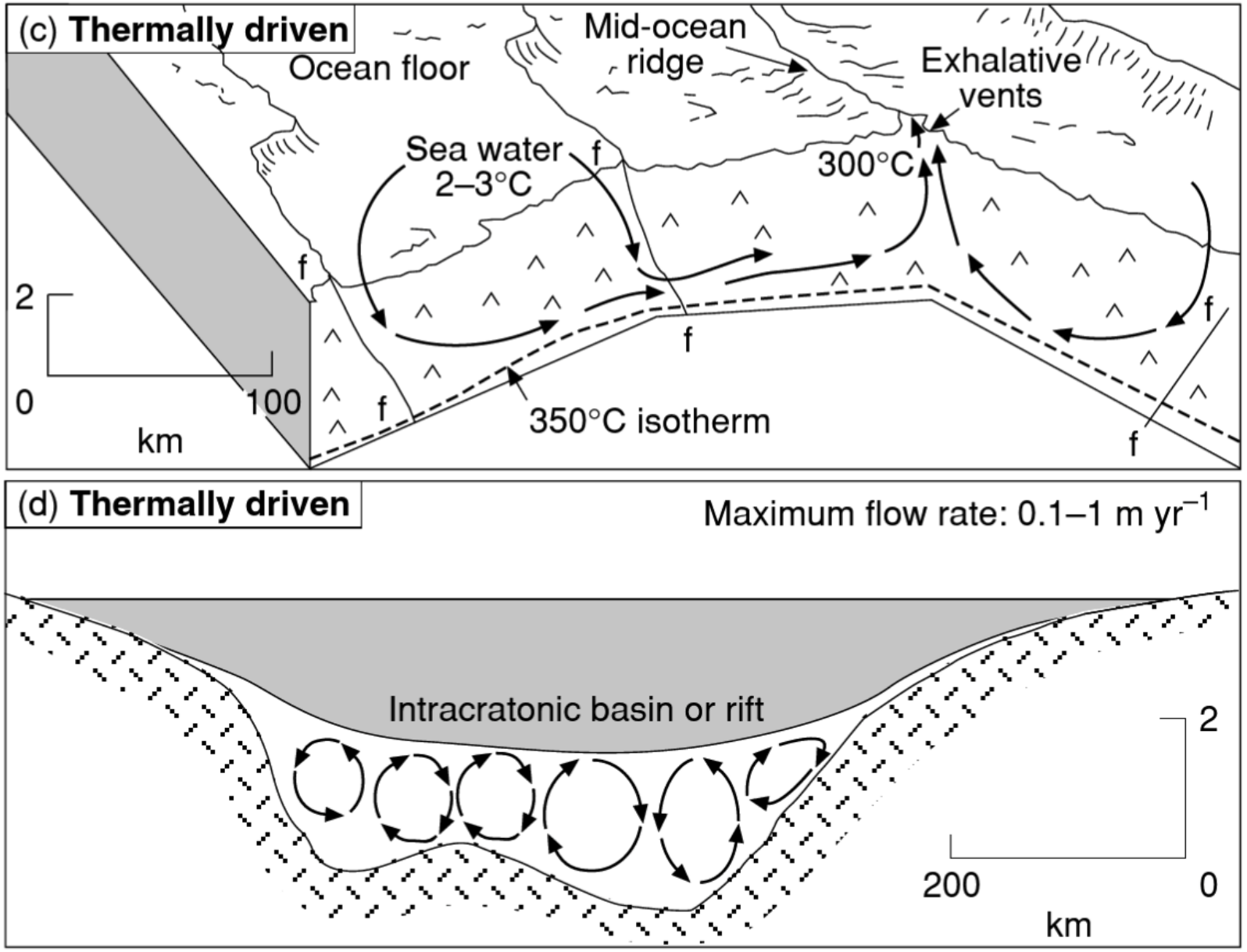
- Brittle deformation in shallow crust via faults, fractures, joints, veins, and cleavage, which allow fluid flow and mineral precipitation

Key Subsurface Properties
Fluid movement



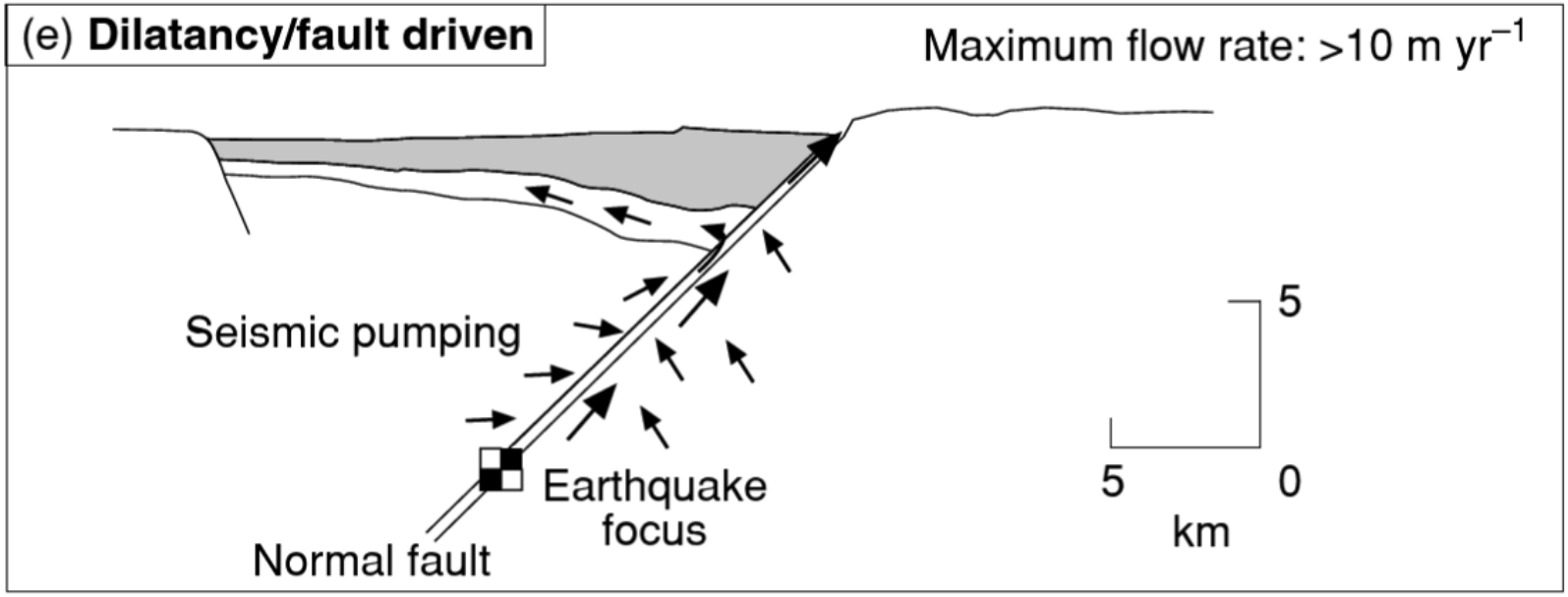
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Key Subsurface Properties
Fluid movement

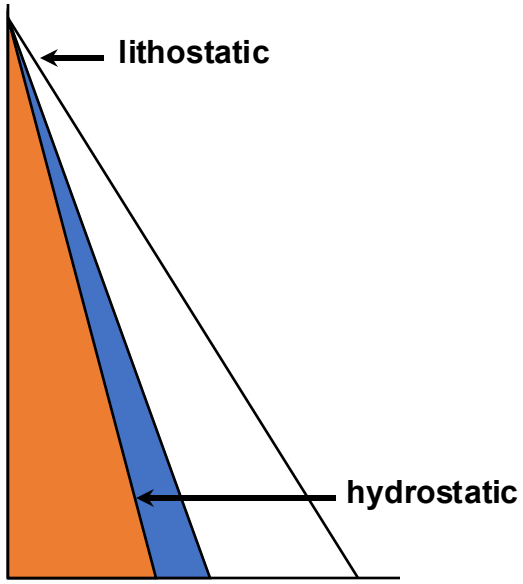


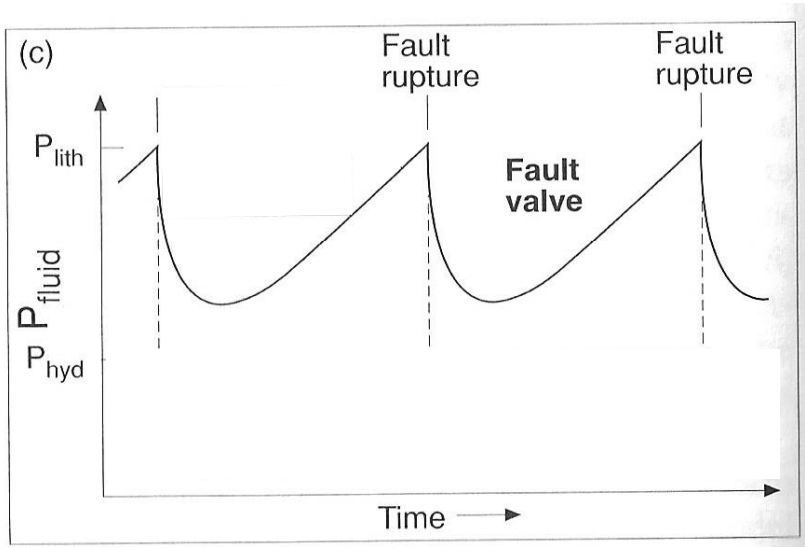
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Key Subsurface Properties
Fluid movement

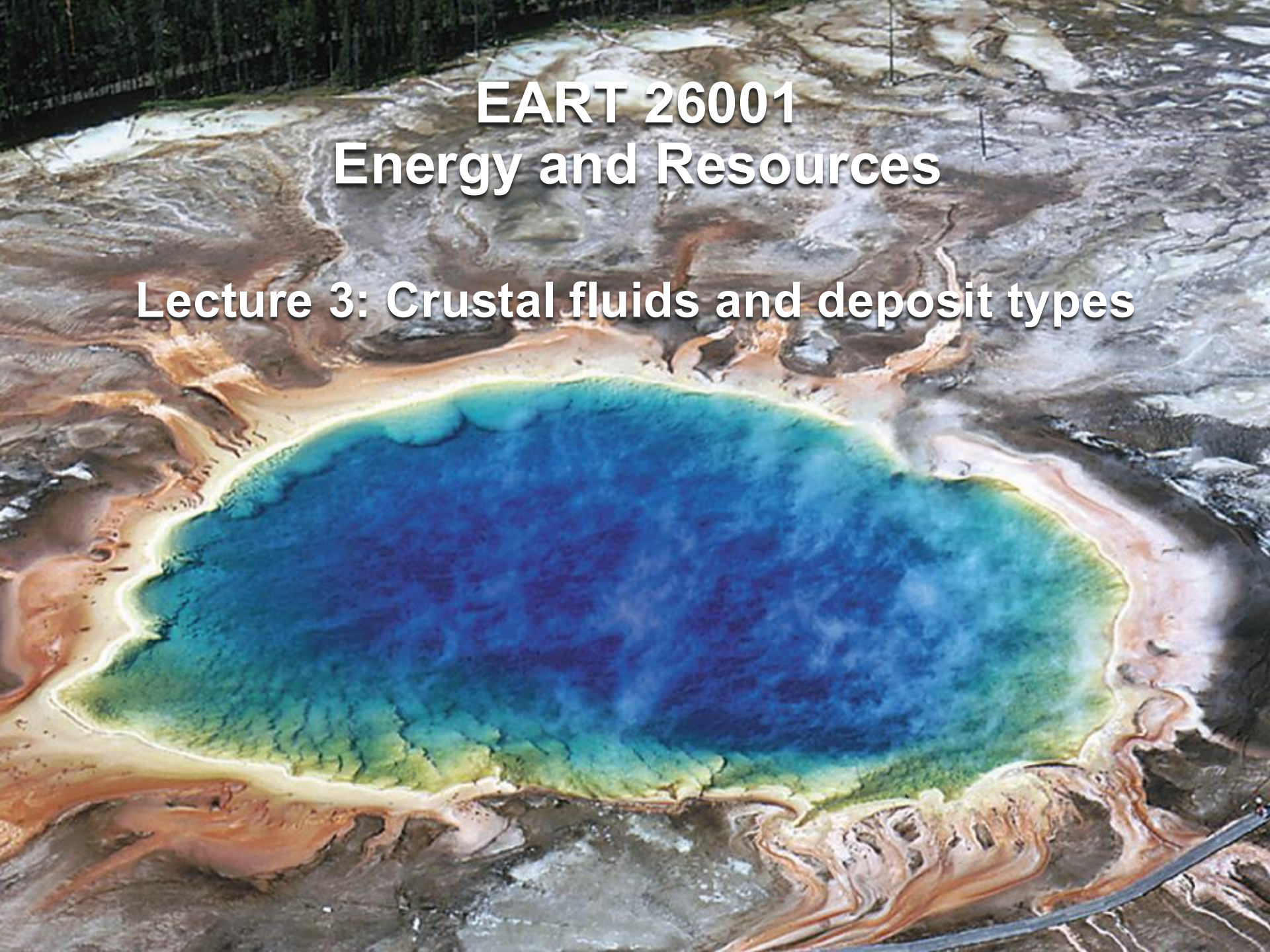


Robb 2005





Robb 2005

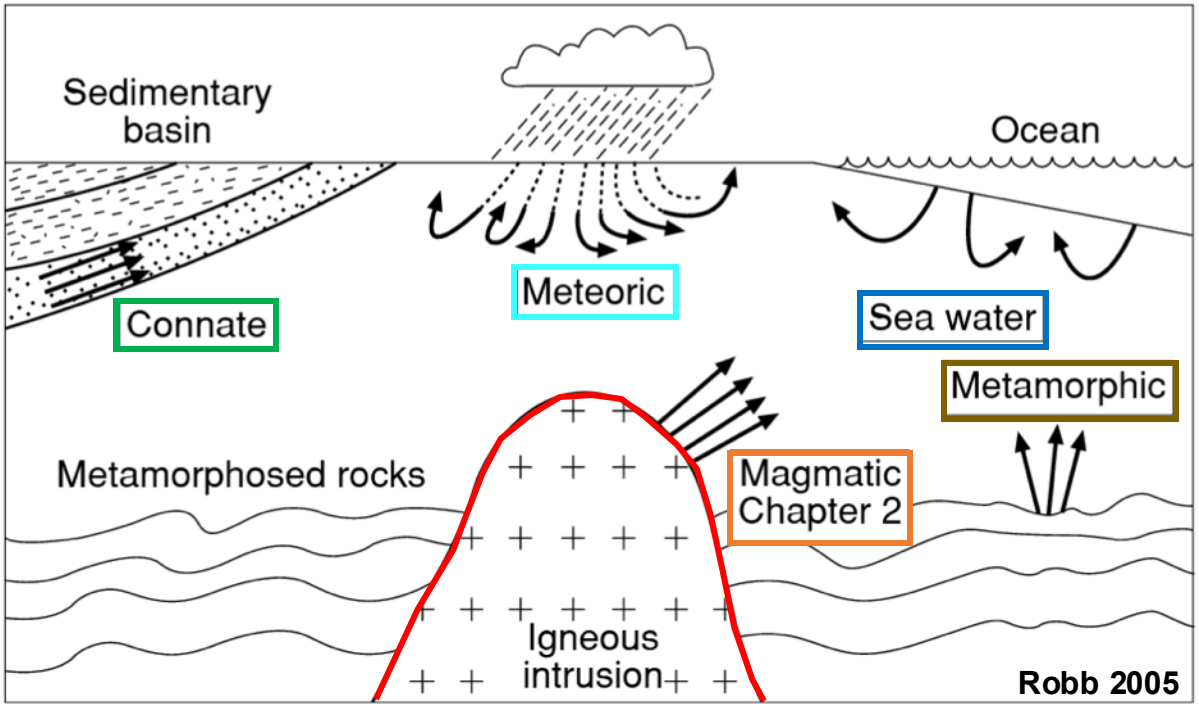


EART 26001
Energy and Resources

Lecture 3: Crustal fluids and deposit types

Introduction
Types of fluids in crust

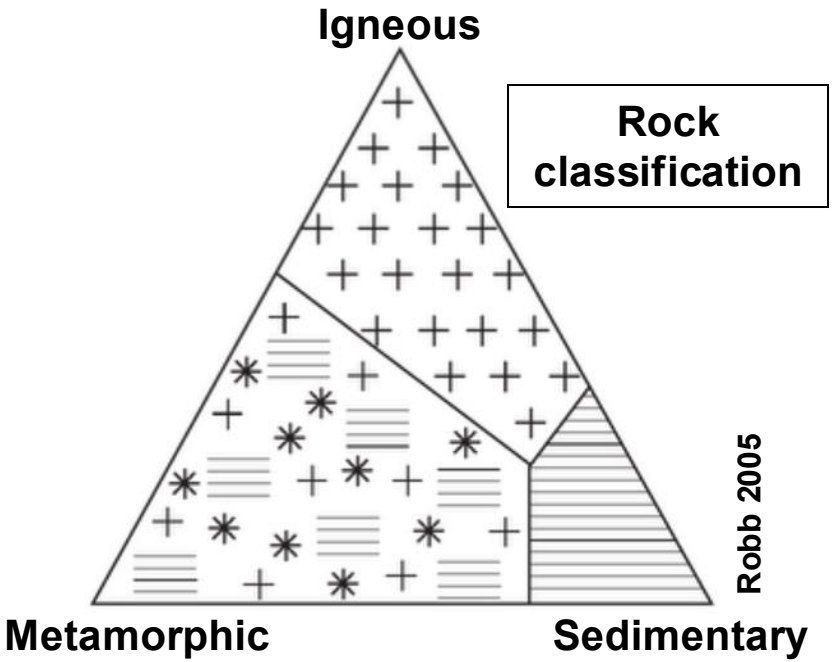
- Magma
- Aqueous fluids:
  - Exsolution from magma
  - Regional metamorphic fluids
  - Connate (formation) water, including saline brines from evaporite dehydration/ dissolution
  - Seawater
  - Meteoric water
- Hydrocarbon fluids

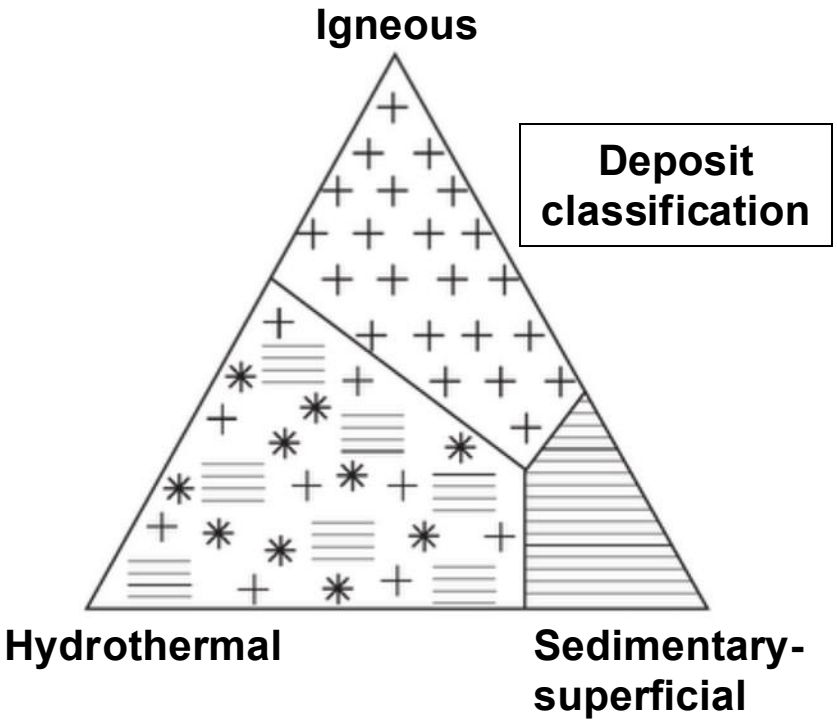


Introduction
Classification of resource deposits

- Igneous/Magmatic
  - Syngenetic igneous processes associated with magma formation, cooling, crystallization
- Magmatic-hydrothermal
  - Epigenetic precipitation from aqueous fluids derived from melt
- Hydrothermal
  - Epigenetic precipitation from hydrothermal aqueous fluids of variable (igneous, metamorphic, surface) origin
- Sedimentary
  - Syngenetic processes during sediment formation
- Surficial and supergene
  - Epigenetic formation from surface fluids
  - Surface alteration of existing deposits

Hydrothermal: thermal energy contained in water
Syngenetic: formed at same time as host rock
Epigenetic: formed after host rock





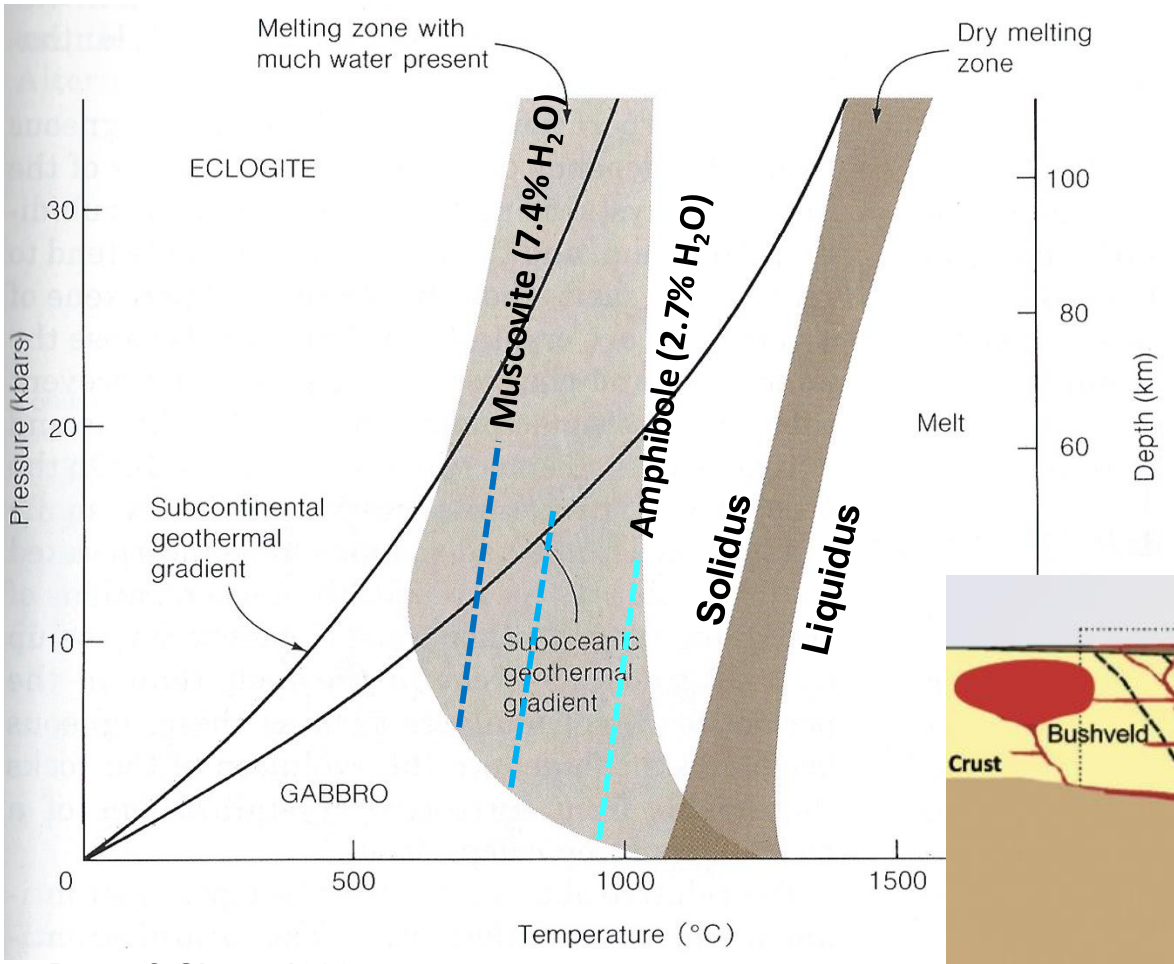
Igneous Deposits and Processes



Photo courtesy M. Polacci

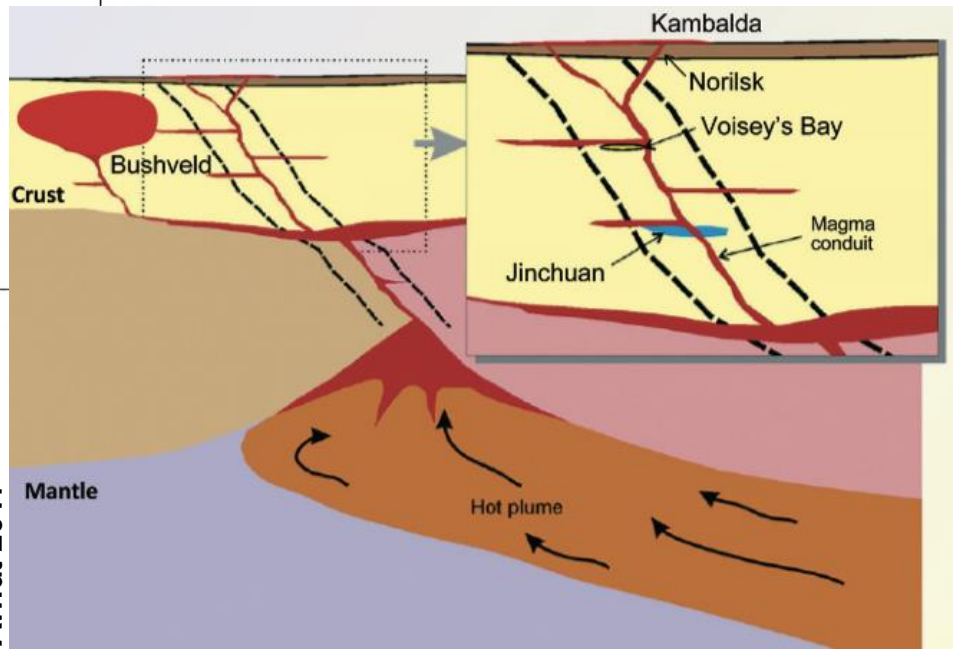
Igneous Deposits and Processes
Magma

Melting of crust



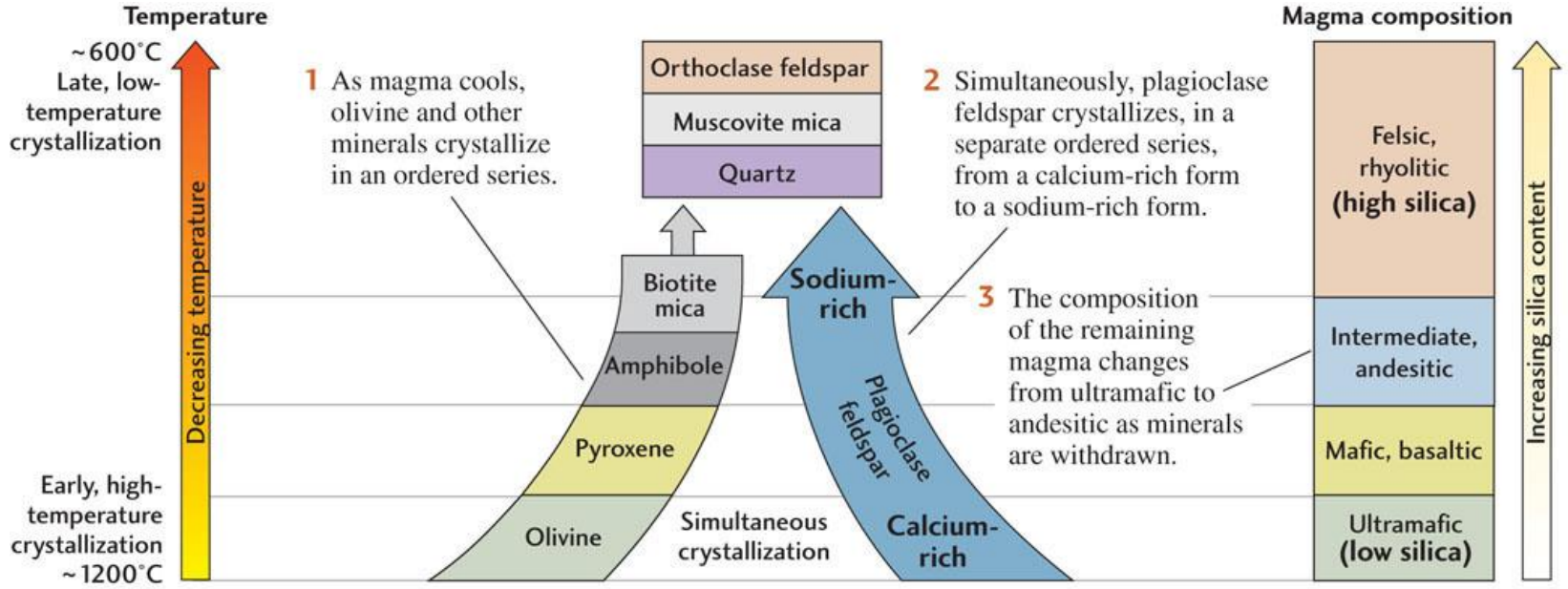
Press & Siever 1986

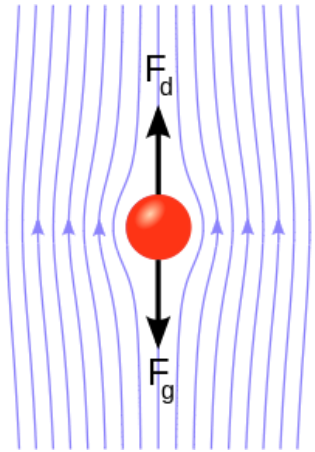
- "normal" geothermal gradient does not intersect mantle solidus
- Partial melting due to:
  - abnormal mantle temperatures – e.g. hot spots, MOR
  - Decompression melting
  - water addition – e.g. subduction zones
  - Orogenic thickening and heating



Arndt 2017

Igneous Deposits and Processes
Crystallization





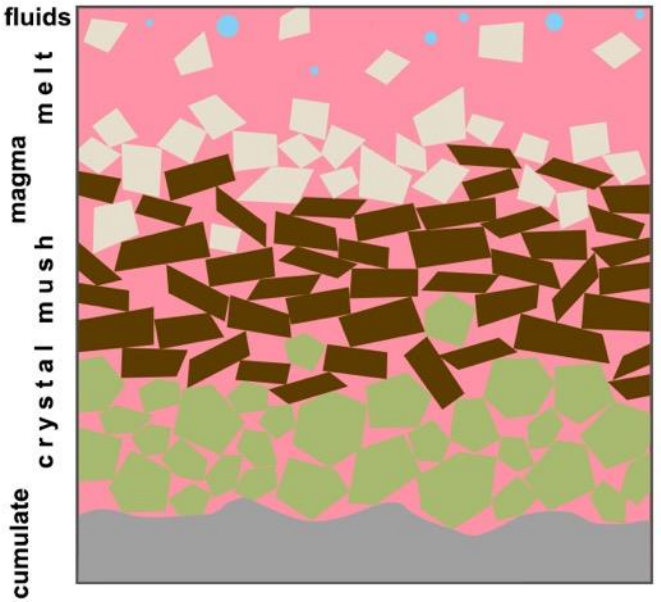
Stoke's Law:

$$V = \frac{(\rho_1 - \rho) \times g \times d^2}{18 \times \mu}$$

Basalt melt:
ρ = 2.65 g/cm^3
μ = 500 poise

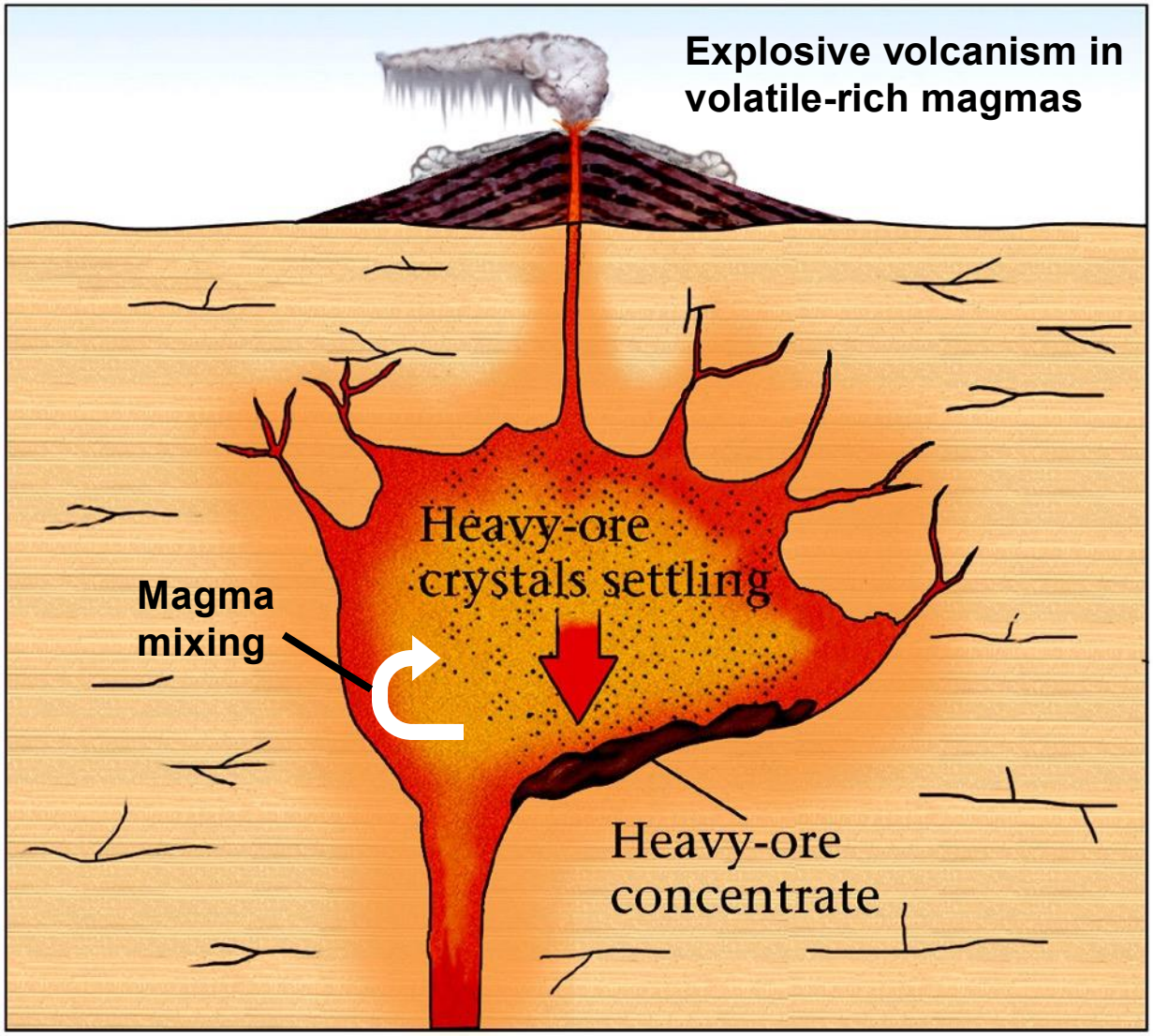
Olivine ρ = 3.3 g/cm^3
Feldspar ρ = 2.6 g/cm^3

For crystals 5 mm:
Olivine v = 1.8 x 10^{-2} cm/s
Feldspar v = 1.4 x 10^{-3} cm/s



© Julien Leuthold - Own work, CC BY-SA 4.0

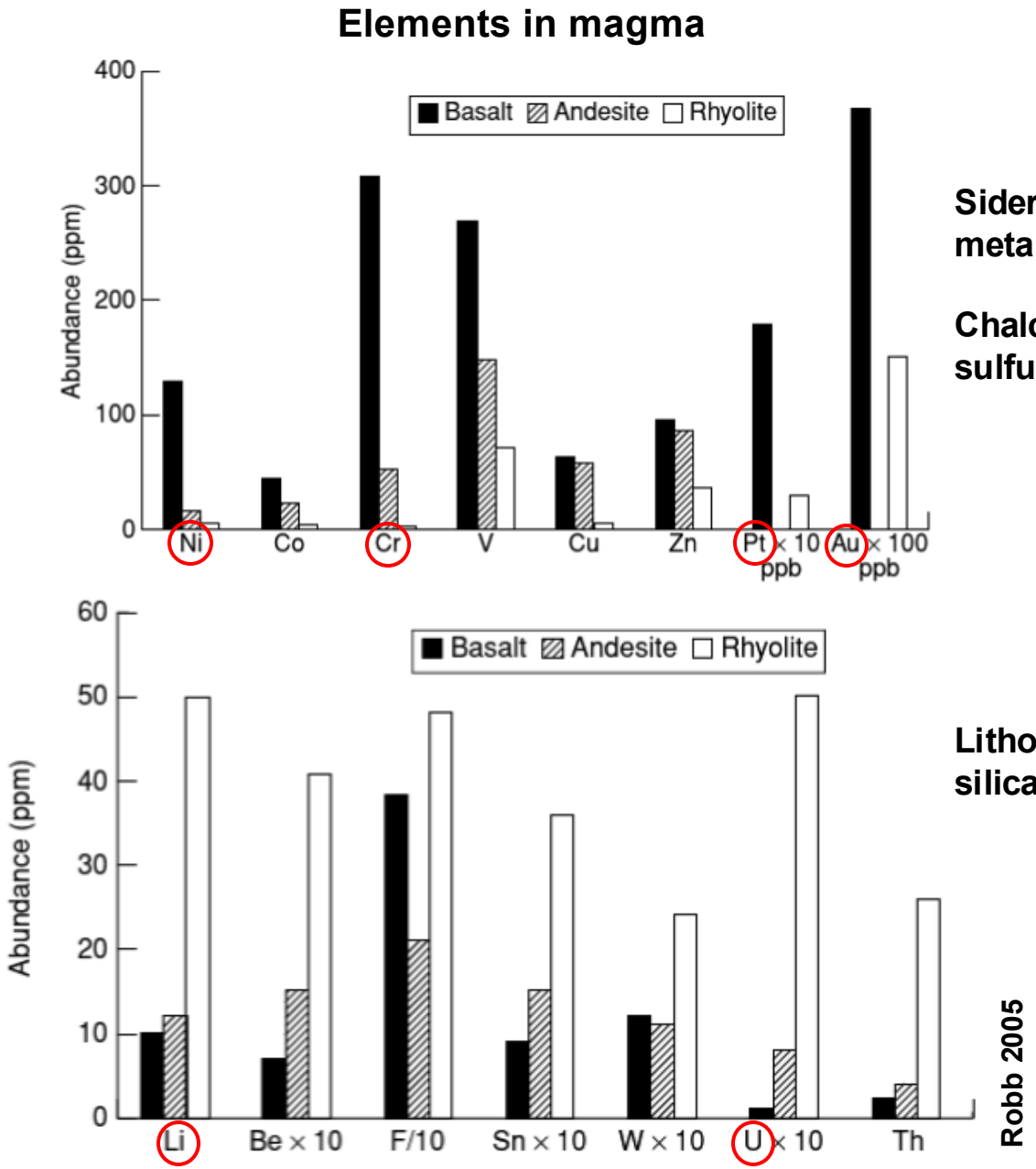
Igneous Deposits and Processes
Magma chamber processes



Earth: Portrait of a Planet, 2nd Edition
Copyright (c) W.W. Norton & Company

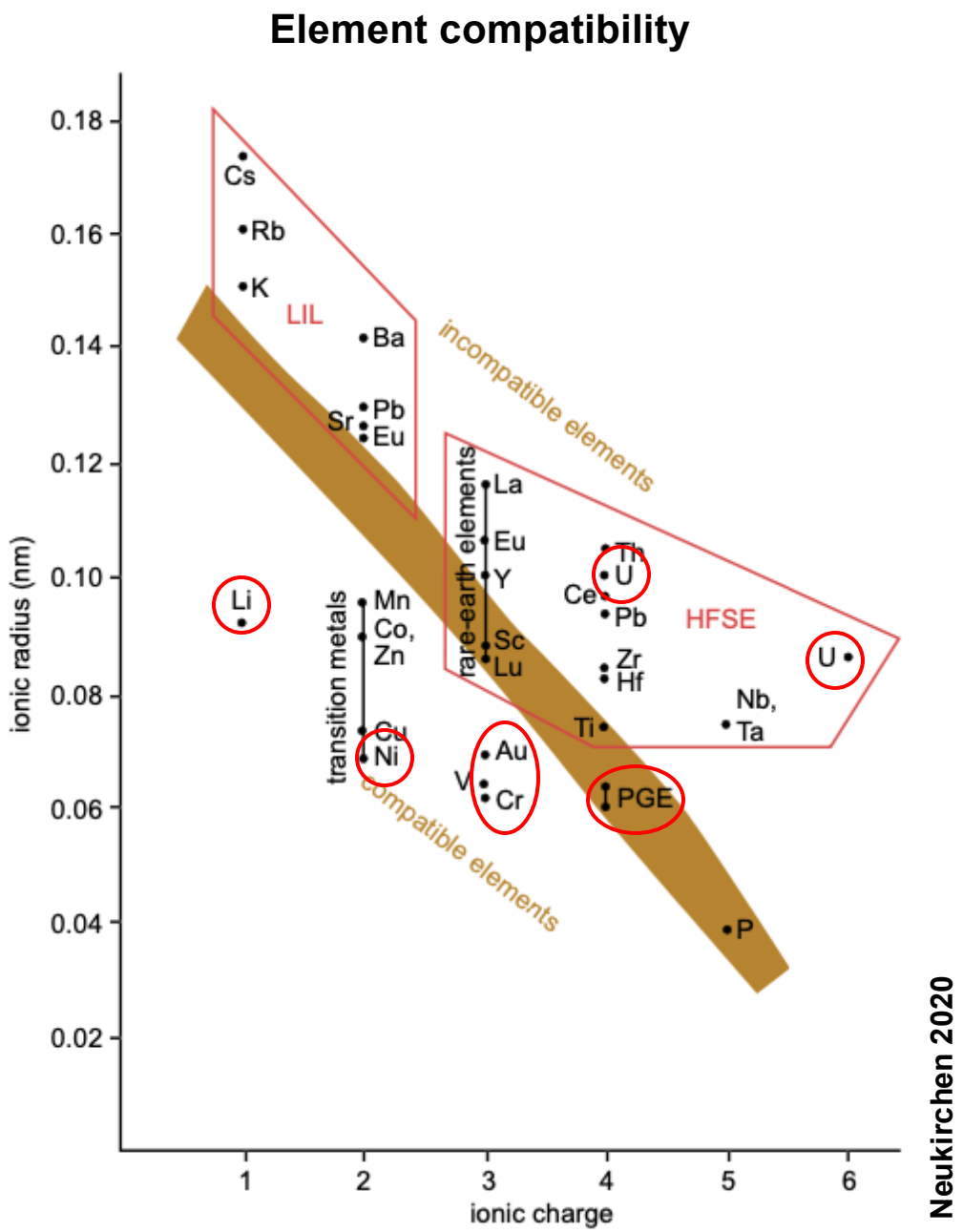
Igneous Deposits and Processes
Element distribution

- Element distribution in igneous rocks and associated resource deposits controlled by element affinity for magma type…..

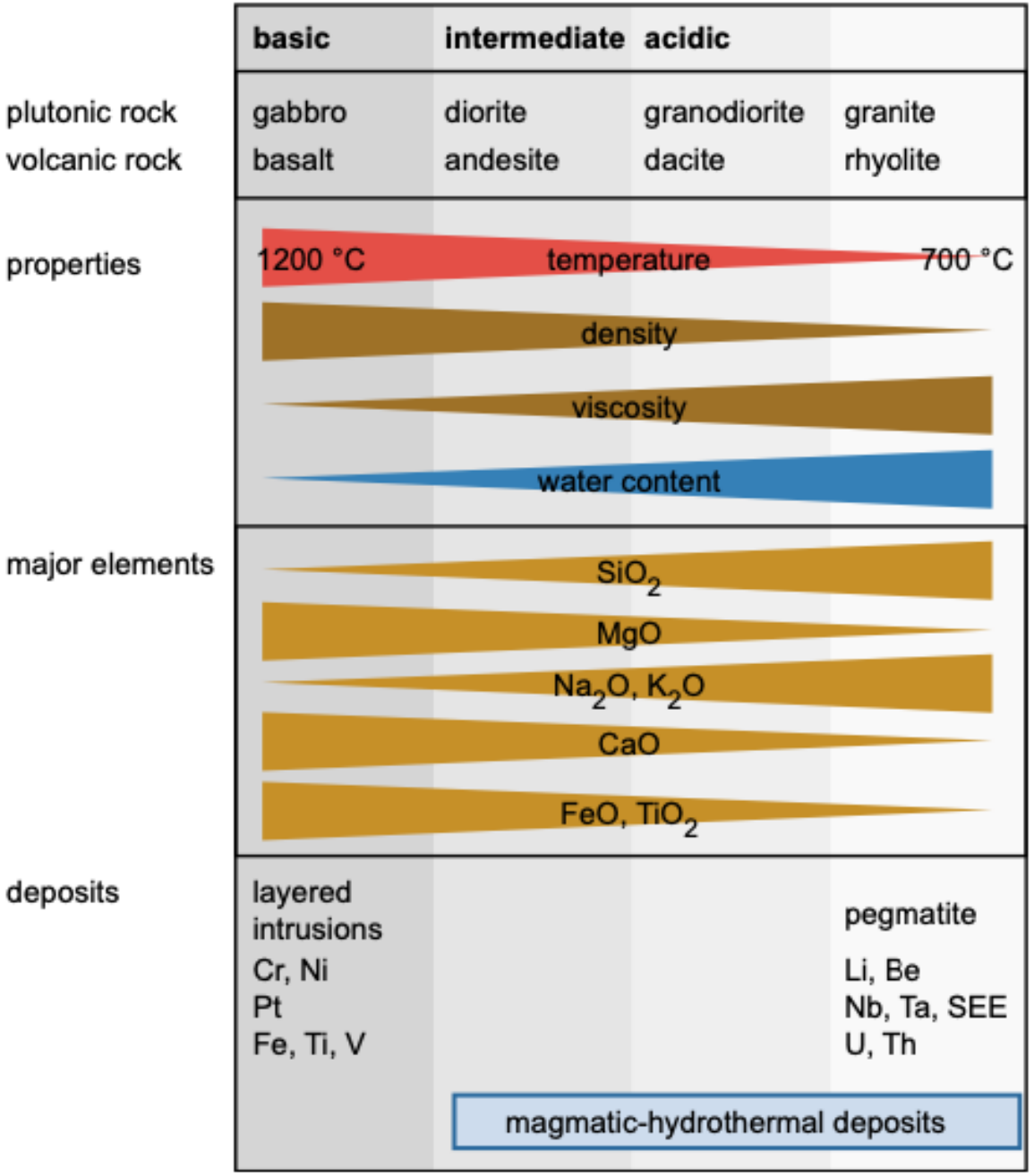


Igneous Deposits and Processes
Element distribution

- …and **element compatibility**



Igneous Deposits and Processes
Key controls



Neukirchen 2020

Igneous Deposits and Processes
Deposit overview

- Explosive volcanism: diamond kimberlites
- Fractionation during crystallization, magma mixing: layered intrusions with Cr, PGE
- Melt immiscibility: Ni-Cu deposits
- Late crystallization and element incompatibility: Sn, U, Th in granites

